Technology Data Sheet (TDS)

TINMORRY TPU 95A

Version 1.0

• Basic Introduction

TINMORRY TPU 95A has ultra-high flexibility and layer strength,so it is resistant to falling, collision, and impact. In addition, it has excellent cold resistance, wear resistance, water resistance and oil resistance, ensuring reliable application under relevant harsh working conditions.

• Specifications

Subjects	Data
Diameter	1.75 ± 0.05 mm
Net filament weight	1000 g ± 0.5%
Length	330 ± 5 m
Spool material	ABS ((Temperature resistance: 80 °C)
Spool size	External Diameter: 198 mm; Inner Diameter: 56 mm; Width: 65 mm

• Recommended Printing Settings

Subjects	Data
Drying settings before printing	75 ± 5 °C，≥ 8 h (Blast drying oven, or filament drying box)
Printing temperature and humidity	≤ 35 ℃, ≤ 20% RH （Put in an air-tight container and sealed with desiccant, and do not dry it with a heating filament drying box during printing, or it can be too soft and cause clogging ）
Storage temperature and humidity	≤ 30 ℃, ≤ 20% RH（Sealed with desiccant）

Compatible support material	Itself or support filament for TPU
Compatible printer type	Open-frame, enclosed-frame (open the door and remove the lad)
Compatible nozzle material	Hardened steel
Compatible nozzle size	0.4 (recommended), 0.6, 0.8 mm
Compatible plate type	Textured PEI Plate or other plate
Plate surface preparation	Glue
Nozzle temperature	210 - 240 °C
Bed temperature	30 - 50 °C (depending on the printer and plate types)
Printing speed*	Max: 250 mm/s; recommended: ≤ 50 mm/s
Chamber temperature	< 35 °C

*When the nozzle temperature = 230 °C, the max printing speed can reach 250 mm/s; however, due to TPU 95A is too soft, when the speed is too high, some thin and tall areas of certain models are prone to shake, resulting in poor printing quality. Therefore, it is recommended to reduce the speed appropriately when printing such models.

• Properties

The commonly used physical properties, mechanical properties, chemical properties, printing properties and other properties of TINMORRY TPU 95A have been tested and are shown in the tables below.

• Physical Properties

Subjects	Testing Methods	Data
Density	ISO 1183	1.21 g/cm^3
Saturated water absorption rate	25 °C , 55% RH	0.98%
Melt index	220 °C, 2.16 kg	26.5 ± 4.1 g/10 min
Glass transition temperature	DSC, 10 °C/min	N / A
Crystallization temperature	DSC, 10 °C/min	N / A
Melting temperature	DSC, 10 °C/min	182 °C
Vicar softening temperature	ISO 306, GB/T 1633	N / A
Heat deflection temperature 1	ISO 75, 1.82 MPa	N / A
Heat deflection temperature 2	ISO 75, 0.45 MPa	N / A

• Mechanical Properties

Properties	Testing Methods	Data
Tensile strength (XY)	ISO 527, GB/T 1040	13 ± 3 MPa
Tensile strength (Z)	ISO 527, GB/T 1040	9 ± 3 MPa
Young's modulus (XY)	ISO 527, GB/T 1040	122 ± 18 MPa
Young's modulus (Z)	ISO 527, GB/T 1040	108 ± 25 MPa
Breaking elongation rate (XY)	ISO 527, GB/T 1040	> 330%
Breaking elongation rate (Z)	ISO 527, GB/T 1040	> 250%
Bending strength (XY)	ISO 178, GB/T 9341	N / A

Bending strength (Z)	ISO 178, GB/T 9341	N / A
Bending modulus (XY)	ISO 178, GB/T 9341	N / A
Bending modulus (Z)	ISO 178, GB/T 9341	N / A
Impact strength (XY)	ISO 179, GB/T 1043	> 124.0 kJ/m²（不断裂）
Impact strength (Z)	ISO 179, GB/T 1043	> 124.0 kJ/m²（不断裂）

• Chemical and Other Properties

Subjects	Data
Component	TPU 95A
Skin irritation	No
Resistance to water	Yes
Resistance to oil and grease	Resistant to most kinds of daily oil and grease
Resistance to organic solvent	Not resistant to some organic solvents
Resistance to weak acid	Yes
Resistance to strong acid	Not resistant
Resistance to weak alkali	Yes
Resistance to strong alkali	Not resistant
Flammability	Yes
Odor during printing	Odorless
Odor during combustion	Pungent

• Specimen Info

Subjects	Data
Size	Shown at the following pictures
Nozzle temperature	220 °C
Bed temperature	35 °C
Printing speed	XY: 50 mm/s; Z: 0 mm/s
Infill density	100%
Infill pattern	Concentric

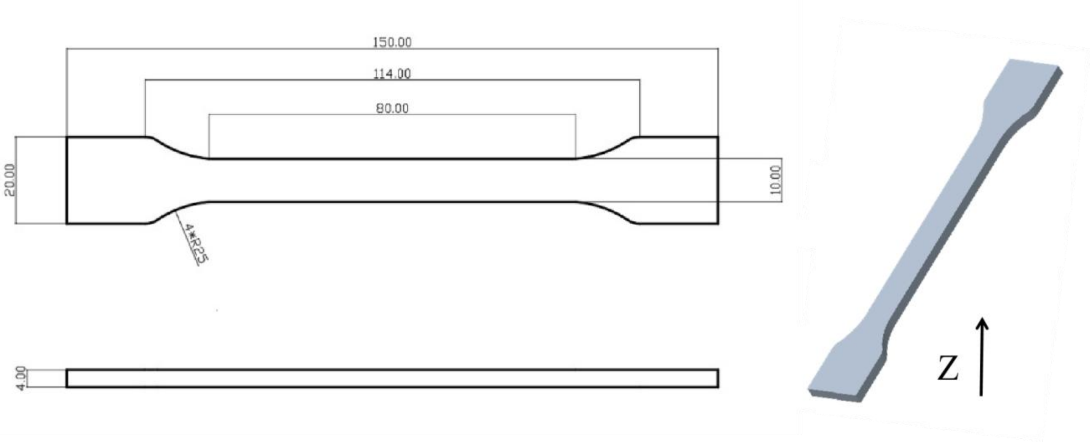
Statements：

1. The test specimens were printed under the key parameters mentioned above. Before testing, all specimens were placed in an indoor environment at about 25 °C for over 12 hours. Please note that when printing, different ambient temperatures, fan speeds and layer times can lead to varying cooling, which can affect the mechanical properties of the specimens, especially those in the Z direction. Additionally, different printers can also result in variations in printing quality.

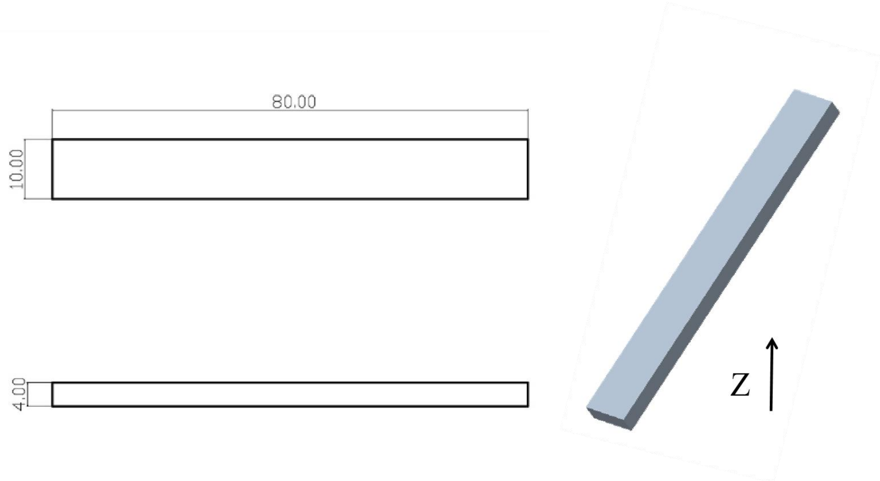
2. If high printing quality is required, it is recommended to dry every kind of TINMORRY filaments before printing and use an air-tight container and desiccant to protect them from moisture during the printing process (with humidity inside the container less than 20%) to reduce the risk of issues related to moisture absorption in filaments. The recommended drying conditions are shown above. Note that when drying them, avoid placing the spools near the heater to prevent damage to the spools or the filaments due to excessive temperature. Do not use a kitchen oven or microwave oven.

3. After annealing, the strength, stiffness, toughness and flexibility of TINMORRY TPU 95A prints can hardly be increased, while some prints may undergo shrinkage or deformation, so annealing is not recommended.

1. Tensile Test



2. Bending Test and Impact Test



• Disclaimer

The above properties data is obtained by TINMORRY through testing with standard samples, standard methods, and specific equipment, and is for reference and comparison only; if some data are changed, TINMORRY will update this TDS, but may not be able to inform the users, and please forgive us. The actual performance of 3D prints depends not only on the performance of the filaments used, but also on many factors, such as the degree of moisture absorption of the filaments, the printer, environmental conditions, model characteristics, printing parameters, etc.. When using TINMORRY FDM 3D printing filaments, users are responsible for the legality, safety, and performance of the prints. TINMORRY is responsible for the quality of our filaments, but not for any damage or injury that occurs during using them,or the usage scenarios of them.